Increment 1 – Test Report

Status: Complete · Oct 24, 2026 · Author: Salam Alsaour
Input: 01_Requirements.md (acceptance tests, Section 6; demo scenario, Section 7; Definition of Done, Section 8), 02_Design.md (test design, Section 11)
Build tested: branch `increment-1`; backend Python 3.11, PostgreSQL 16; frontend Node 22, React 18, Vite 8

1. Summary

Check	Result
Backend tests (pytest, real PostgreSQL)	**108 passed**, 0 failed
Backend line coverage	**95%** of `app` (gate 70%; seed, CLI, scheduler wiring, and app factory excluded by configuration)
Permission matrix	107 routes listed; **494** calls by callers that are not allowed, all answered `401` or `403`
Acceptance tests of 01_Requirements Section 6	**43 of 43 passed** (automated, plus the browser checks of AT-AUT-01, AT-PRF-05, and AT-LOC-01)
Non-functional checks of 01_Requirements Section 4.2	14 of 14 met; NFR-MNT-01 is confirmed when CI runs on the pull request
Frontend	TypeScript strict build clean; **7 Vitest tests passed** (translation keys and date format); production build succeeds
Lint	`ruff` clean on `app`, `tests`, `alembic`, `scripts`
M3 demo dry run on `docker compose up`	**10 of 10 steps passed**, twice in a row, with real emails through Mailpit
Dependencies shipped to users (`npm audit --omit=dev`)	0 known vulnerabilities
Committed scope	39 Must requirements delivered; 4 of 4 stretch items delivered (FR-ORG-09, FR-CYC-05, FR-VAC-05, FR-VAC-08); FR-ORG-07 deferred to Increment 3 as planned

2. How to Repeat the Tests

```
docker compose up -d db mailpit          # PostgreSQL on 5433, Mailpit on 8025
cd backend
pip install -r requirements-dev.txt
ruff check app tests alembic scripts
pytest --cov=app                         # creates the internsync_test database on first use
cd ../frontend
npm ci
npx tsc -b && npx vitest run && npx vite build
```

The M3 rehearsal against the full stack:

```
docker compose up -d --build
docker compose exec api python -m app.seed --demo
python backend/scripts/demo_dry_run.py   # needs httpx (in requirements-dev.txt)
```

3. Acceptance Tests

Every test of 01_Requirements Section 6 has a named test in `backend/tests` (the AT ID is in the function name). "Browser" means the check was also done by hand in the running app, in Arabic and English.

Test	Automated test(s)	Result
AT-AUT-01	`test_at_aut_01_each_role_logs_in_with_its_role`; browser: each role's menu, the team page only for the Company Admin	Pass
AT-AUT-02	`test_at_aut_02_registration_kinds_and_single_use_confirmation`, `test_at_aut_02_expired_confirmation_link`, `test_resend_confirmation`	Pass
AT-AUT-03	`test_at_aut_03_idle_timeout_and_logout`, `test_session_max_lifetime`	Pass
AT-AUT-04	`test_at_aut_04_password_reset`, `test_at_aut_04_expired_reset_link`	Pass
AT-AUT-05	`test_every_route_is_in_the_matrix_and_nothing_else`, `test_at_aut_05_disallowed_callers_are_refused`, `test_allowed_callers_pass_the_role_gates`; scope cases in the module suites	Pass
AT-AUT-06	`test_at_aut_06_deactivation`, `test_deactivated_user_session_ends`, `test_company_suspend_and_reactivate`	Pass
AT-ORG-01	`test_at_org_01_new_college_appears_and_deactivated_does_not`, `test_registration_refused_for_inactive_college`	Pass
AT-ORG-02	`test_at_org_02_unverified_accounts_are_limited`	Pass
AT-ORG-03	`test_at_org_03_one_coordinator_per_college`	Pass
AT-ORG-04	`test_at_org_04_company_verification`	Pass
AT-ORG-05	`test_at_org_05_student_verification_enrols`, `test_student_request_waits_without_open_cycle`	Pass
AT-ORG-06	`test_at_org_06_invited_supervisor_joins_the_college`, `test_at_org_06_expired_invitation_and_resend`, `test_company_admin_invites_colleague`	Pass
AT-ORG-08	`test_at_org_08_document_access_and_purge` (clock moved 29 and 31 days)	Pass
AT-ORG-09	`test_at_org_09_resubmission_after_rejection`	Pass
AT-CYC-01	`test_at_cyc_01_unit_weeks_follow_weekday_and_time`, `test_at_cyc_01_api_creates_twelve_weeks`, `test_unit_partial_last_week_counts_only_days_in_the_cycle`	Pass

Test	Automated test(s)	Result
AT-CYC-02	`test_at_cyc_02_weights_must_total_100`	Pass
AT-CYC-03	`test_at_cyc_03_locked_rubric_cannot_change`	Pass
AT-CYC-04	`test_at_cyc_04_unit_synonym_matches_and_shows_canonical_name`, `test_skill_catalog_admin`	Pass
AT-CYC-05	`test_at_cyc_05_closed_cycle_is_read_only`, `test_closed_cycle_freezes_reports`	Pass
AT-PRF-02	`test_at_prf_02_verified_student_cannot_edit_academic_fields`, `test_pending_student_can_correct_academic_fields`	Pass
AT-PRF-03	`test_at_prf_03_size_and_type_limits_and_versions`	Pass
AT-PRF-04	`test_at_prf_04_unit_full_dictionary_on_sample_cvs`, `test_at_prf_04_extracted_skills_are_only_candidates`, `test_arabic_docx_cv_extraction`	Pass
AT-PRF-05	`test_at_prf_05_image_only_pdf`; browser: the message and the empty editable list	Pass
AT-VAC-01	`test_at_vac_01_create_and_validate`	Pass
AT-VAC-02	`test_at_vac_02_visibility`	Pass
AT-VAC-03	`test_at_vac_03_filters`	Pass
AT-VAC-04	`test_at_vac_04_application_rules` (duplicate, wrong training type, after deadline, sixth application), `test_unit_eligibility_reasons`	Pass
AT-VAC-05	`test_at_vac_05_withdraw_frees_a_slot`	Pass
AT-VAC-06	`test_at_vac_06_applicants_and_decisions`	Pass
AT-VAC-07	`test_at_vac_07_placement_rules`, `test_placement_with_another_company_supervisor`	Pass
AT-VAC-08	`test_at_vac_08_change_supervisor_and_cancel`	Pass
AT-REP-01	`test_at_rep_01_slots_and_deadlines`	Pass

Test	Automated test(s)	Result
AT-REP-02	`test_at_rep_02_attachment_limits`, `test_weekly_report_needs_all_fields`, `test_final_report_needs_a_main_document`	Pass
AT-REP-03	`test_at_rep_03_drafts_private_and_submission_final`	Pass
AT-REP-04	`test_at_rep_04_unit_deadline_minute_is_inclusive`, `test_at_rep_04_on_time_late_and_missing`	Pass
AT-REP-05	`test_at_rep_05_revision_cycle_keeps_versions`	Pass
AT-REP-06	`test_at_rep_06_comments`	Pass
AT-REP-07	`test_at_rep_07_closed_slot_refuses_submission`	Pass
AT-WFL-01	`test_at_wfl_01_every_allowed_transition_passes`, `test_at_wfl_01_disallowed_transitions_from_each_state_are_rejected`	Pass
AT-LOC-01	`test_language_is_saved_and_used_for_errors`; browser: every Increment 1 page in both languages, right to left in Arabic and left to right in English	Pass
AT-LOC-02	`frontend/src/i18n/keys.test.ts` (same keys in both files, every key used in the code exists, placeholders match); 506 keys	Pass
AT-AUD-01	`test_m3_demo_scenario_and_at_aud_01`; dry-run step 10	Pass
AT-AUD-02	`test_at_aud_02_no_endpoint_changes_audit_entries`, `test_at_aud_02_database_refuses_update_and_delete`, `test_at_aud_02_filters_and_scope`	Pass

4. Non-Functional Checks

ID	Evidence	Result
NFR-SEC-02	Argon2id hashing (`app/core/security.py`); `test_weak_password_refused`	Met
NFR-SEC-03	`test_lockout_after_five_failures` (locked, then open again after 16 minutes); registration limit `test_registration_rate_limit`	Met
NFR-SEC-04	Type detected from content (`test_at_prf_03…` refuses a PNG named `.pdf`, a ZIP named `.docx`, and text); files stored under random keys outside the web root; `403` without permission (`test_at_org_08…`, `test_cv_file_visible_to_company_only_after_application`); downloads send `nosniff`	Met
NFR-PRV-01	`test_consent_gate`; privacy notice page at first login	Met
NFR-PRV-02	Text extraction and skill matching run inside the API (`app/ai`); no outside service is called	Met

ID	Evidence	Result
NFR-PRV-03	Permission matrix; the Platform Admin is refused on CV, report, and placement routes; audit entries of the college only for its coordinator (`test_at_aud_02_filters_and_scope`)	Met
NFR-REL-04	A refused attachment (too large or wrong type) leaves the draft and its other files unchanged (`test_at_rep_02_attachment_limits`)	Met
NFR-L10N-01	Browser check of every page in both languages and directions	Met
NFR-L10N-02	Deadlines stored in UTC and shown in Asia/Riyadh as DD/MM/YYYY HH:mm (`format.test.ts`); 1 minute before is on time, 1 minute after is late (`test_at_rep_04_unit_deadline_minute_is_inclusive`)	Met
NFR-MNT-01	`.github/workflows/ci.yml` runs lint, tests with the 70% gate, and the frontend checks on every pull request; local coverage 95%	Met locally; CI run on the pull request
NFR-MNT-02	OpenAPI generated by FastAPI; the frontend types are generated from it, and CI fails if either committed copy is out of date	Met
NFR-MNT-03	Link validity, session timeout, lockout, application limit, file sizes, attachment count, and purge days are read from `system_settings` (`test_majors_and_settings`, `test_at_vac_05`… changes the limit)	Met
NFR-PRT-02	`docker compose up -d --build` starts database, API (migrations and reference data on start), web app, and Mailpit; the dry run ran against it	Met
NFR-USA-03	Checked at 375 px (phone, the sidebar stacks above the content) and 1280 px	Met

5. M3 Demo Dry Run

Run with `backend/scripts/demo_dry_run.py` against `docker compose up` on Oct 24, 2026. Emails were read from Mailpit, and the demo clock was moved past the week 2 deadline for step 8.

Step	Result
1. Platform Admin adds a college and a skill	Pass
2. Coordinator registers with an appointment letter, confirms the email, is approved	Pass
3. 12-week Co-op cycle with 12 deadlines, weights 40/30/30, a rubric; invited academic supervisor joins	Pass
4. Company registers and is approved; Company Admin invites a colleague; vacancy published for the university	Pass
5. Student registers, sees the pending status, completes the profile, uploads a CV (6 skills suggested), confirms the skills	Pass
6. Coordinator approves and enrols the student; the filtered list shows the vacancy; the student applies	Pass
7. Company accepts; coordinator confirms the placement; 13 report slots appear	Pass
8. Week 1 submitted late, revision requested, resubmitted (2 versions kept), accepted; week 2 closed as Missing	Pass
9. Language switched to English and back, saved on the account	Pass
10. Coordinator's audit log shows the student's 14 entries covering every step	Pass

The same steps were also done by hand in the browser for the student (writing and submitting a report) and the academic supervisor (requesting a revision), and every page of the coordinator, company, and Platform Admin was opened in both languages.

6. Defects Found and Fixed

Defect	Found by	Fix
Sessions ended at once when the server's clock and the application clock differed (token issue time checked by the JWT library)	`test_at_aut_01…`	Expiry checked against the application clock only
Error messages ignored the user's saved language	`test_language_is_saved_and_used_for_errors`	Language kept on the request and applied when the error is written
Publishing a vacancy in the request that created it failed	Demo seed	Database session flushes pending rows before queries
A failed job could leave its lock held on a pooled connection, so the job never ran again	`test_job_failure_is_recorded_and_releases_the_lock`	Transaction-level advisory lock
The API container stopped at start with a plain `CORS_ORIGINS` value	`docker compose up`	Comma-separated values accepted; `test_config.py`
The Missing flag waited 15 minutes after every restart	Dry run on Docker	First run 30 seconds after start
Fractions such as `3.10 / 5.00` were reversed on Arabic pages	Browser check	Shown left to right
Single-word synonyms such as "rest", "spring", and "العربية" matched ordinary CV text	Review of the skill vocabulary	Removed; "Go" and "R" renamed

7. Known Limitations and Follow-Up

- **Development tooling advisories:** `npm audit` reports 3 moderate advisories in Tailwind CSS 3 and its PostCSS helpers, which run only while the site is built. The fix is the move to Tailwind 4, planned for Increment 2. Vite and Vitest were upgraded in this increment for the same reason.
- **Backend type checking:** the frontend is type-checked in strict mode; the backend relies on `ruff` and the tests. A type checker for the backend is planned for Increment 2.
- **Browser tests:** the end-to-end checks of this increment were done by hand. Automated browser tests are planned with the Increment 2 screens.
- **Skill extraction accuracy** is measured at M4 on labelled CVs (decision RQ-4).
- **Performance and HTTPS** are checked on the free-tier deployment at the start of Increment 2 (decision RQ-7).
- **Email delivery** is tested through Mailpit; the real SMTP account is configured with the deployment.

8. Definition of Done

Item (01_Requirements Section 8)	Status
Code merged into `main` through a pull request with CI green	Pull request opened from `increment-1`; merge after CI passes
Acceptance tests automated and passing; endpoints in the permission matrix	Done (Sections 3 and 1)
Screens in Arabic and English following the wireframes	Done

Item (01_Requirements Section 8)	Status
Actions write audit entries	Done (AT-AUD-01)
Design, API, or SRS updated where the implementation changed something	Done: CR-07, Design 1.3, API 1.2, 02_Design.md Section 12
Demo scenario runs without help	Done (Section 5)

Increment 2 – Test Report

Status: Complete · Oct 26, 2026 · Author: Salam Alsaour
Input: 01_Requirements.md (acceptance tests, Section 6; demo scenario, Section 7; Definition of Done, Section 8), 02_Design.md (test design, Section 11; changes, Section 12)
Build tested: branch `increment-2`; backend Python 3.11, PostgreSQL 16 with pgvector; embedding model `intfloat/multilingual-e5-small@614241f622f5`; frontend Node 22, React 18, Vite 8

1. Summary

Check	Result
Backend tests (pytest, real PostgreSQL)	**158 passed**, 0 failed (108 at the end of Increment 1)
Backend line coverage	**95%** of `app` (5,190 statements; gate 70%)
Permission matrix	129 routes listed (22 new); **627** calls by callers that are not allowed, all answered `401` or `403`
Acceptance tests of 01_Requirements Section 6	**27 of 27 passed** (automated; AT-LOC-03 also checked in the browser)
Non-functional checks of 01_Requirements Section 4.2	**8 of 8 met**
AI evaluation (`ai-eval/evaluate.py`)	Skill extraction recall **97.2%** (target 70%); top-5 hit rate **100%** (target 80%). The set is easy; see Section 5 for what that number does and does not show
Frontend	TypeScript strict build clean; **8 Vitest tests passed**; production build succeeds; 648 translation keys in each language
Lint	`ruff` clean on `app`, `tests`, `alembic`, `scripts`
M4 demo scenario	Steps 1–9 automated end to end through the HTTP API (`test_m4_demo_scenario`); step 10 run; the screens walked through in the browser
Regression on the full stack	`docker compose up` with the new image: the M3 dry run passed **10 of 10** steps; the student's ranking ran with the model
Deployment parts	Image built with the model; started as the Space starts it (production settings, port 7860): model loaded offline in about 15 s, jobs ran; backup and restore rehearsed
Dependencies shipped to users (`npm audit --omit=dev`)	0 known vulnerabilities
Committed scope	24 of 24 requirements: 18 Must, 4 committed Should, FR-MAT-05 pulled forward, and FR-MAT-06 (stretch) delivered. The three stretch follow-ups from Increment 1 were not reached (Section 9)

2. How to Repeat the Tests

```
docker compose up -d db mailpit
cd backend
pip install -r requirements-dev.txt
ruff check app tests alembic scripts
pytest --cov=app                          # 158 tests; the tests use a stand-in for the model
cd ../frontend
npm ci
npx tsc -b && npx vitest run && npx vite build
```

The AI evaluation needs the model packages (`pip install -r backend/requirements-ai.txt`; the model is downloaded once):

```
python ai-eval/evaluate.py                # with the model
python ai-eval/evaluate.py --basic        # rules only
```

The full stack and the M3 rehearsal:

```
docker compose up -d --build
docker compose exec api python -m app.seed --demo
python backend/scripts/demo_dry_run.py
```

3. Acceptance Tests

Every test of 01_Requirements Section 6 has a named test in `backend/tests`; the AT ID is in the function name.

Test	Automated test(s)	Result
AT-MAT-01	`test_at_mat_01_ranking_holds_eligible_vacancies_in_descending_fit`; evaluation script (Section 5)	Pass
AT-MAT-02	`test_at_mat_02_unit_required_skills_raise_the_score`	Pass
AT-MAT-03	`test_at_mat_03_gpa_below_minimum_is_left_out_and_explained`	Pass
AT-MAT-04	`test_at_mat_04_unit_basic_mode_without_a_vector`, `test_at_mat_04_basic_mode_without_the_model`, `test_vacancy_without_embedding_is_scored_by_the_rules`; the timed test in basic mode	Pass
AT-MAT-05	`test_at_mat_05_unit_matched_and_missing_are_the_set_comparison`; the API lists in `test_at_mat_01…`	Pass
AT-MAT-07	`test_at_mat_07_ranking_is_advisory_and_any_eligible_vacancy_can_be_applied_to`	Pass
AT-ATT-01	`test_at_att_01_full_attendance_in_one_action`	Pass
AT-ATT-02	`test_at_att_02_locked_after_seven_days_until_unlocked`	Pass
AT-ATT-03	`test_at_att_03_percentage_and_visibility`	Pass
AT-ATT-04	`test_at_att_04_unrecorded_week_on_the_company_dashboard_on_day_4`	Pass
AT-EVL-01	`test_at_evl_01_all_criteria_and_comment_required`	Pass

Test	Automated test(s)	Result
AT-EVL-02	`test_at_evl_02_only_the_cycle_rubric`	Pass
AT-EVL-03	`test_at_evl_03_student_sees_periodic_now_and_final_when_the_grade_is_final`	Pass
AT-EVL-04	`test_at_evl_04_read_only_after_submit_and_reopened_once`, `test_final_evaluation_cannot_be_reopened_while_the_grade_is_submitted`	Pass
AT-WFL-02	`test_at_wfl_02_unit_total_rounds_half_up`, `test_at_wfl_02_grade_equals_the_hand_calculation`	Pass
AT-WFL-03	`test_at_wfl_03_submit_refused_with_the_missing_items`	Pass
AT-WFL-04	`test_at_wfl_04_and_06_return_approve_and_what_the_student_sees`	Pass
AT-WFL-05	`test_at_wfl_05_final_grade_locked_and_reopened_with_a_reason`	Pass
AT-WFL-06	`test_at_wfl_04_and_06_…`, `test_grade_visibility`	Pass
AT-WFL-07	`test_at_wfl_07_bulk_approval_one_audit_entry_each` (10 grades, 10 audit entries)	Pass
AT-DSH-01	`test_at_dsh_01_every_count_equals_the_database`	Pass
AT-DSH-02	`test_at_dsh_02_cycle_figures_overall_and_per_company`	Pass
AT-DSH-05	`test_at_dsh_05_platform_figures`	Pass
AT-PER-02	`test_at_per_02_ranking_1000_vacancies_within_5_seconds` (Section 4)	Pass
AT-AI-02	`test_application_keeps_the_fit_snapshot`, `test_version_and_health_name_the_loaded_model`	Pass
AT-AI-03	`ai-eval/evaluate.py` (Section 5)	Pass
AT-LOC-03	`frontend/src/i18n/keys.test.ts` (648 keys, same in both files, every literal key exists, placeholders match); browser check of every new screen in both languages	Pass

4. Non-Functional Checks

ID	Evidence	Result
NFR-PER-02	1,000 published vacancies, 20 ranking requests per mode: median 0.69 s and 95th percentile **0.83 s** with the embeddings, 0.68 s and **0.76 s** in basic mode (limit 5 s)	Met
NFR-PER-03	`test_nfr_per_03_skills_from_a_5_mb_cv_within_30_seconds`: a 4.8 MB PDF packed with text is read and its skills suggested in about 2–6 s (limit 30 s). Before the fix in Section 7 the same file took 94 s	Met
NFR-PRV-02	The model runs inside the API process; the image holds it and sets `HF_HUB_OFFLINE=1`. In the container started as on the Space, the log shows no request to the Hugging Face Hub	Met
NFR-AI-01	Every ranking carries `advisory: true`, the parts, and the matched and missing skills, shown under the Advisory label; AT-MAT-07 shows that no application changes without a person	Met
NFR-AI-02	`match_scores` and each application snapshot store `model_version`; `/version` and the deep health check name the loaded model	Met
NFR-AI-03	`ai-eval/data` (20 profiles, 30 vacancies) and `ai-eval/evaluate.py` are in the repository; three runs on Oct 25–26 gave the same numbers	Met
NFR-L10N-01	Browser check of every new screen in Arabic (right to left) and English (left to right): dashboards of the five roles, recommended vacancies and the plan, the placement tabs (reports, attendance, evaluations, grade), the grades page, cycle statistics, platform figures, the new settings	Met
NFR-USA-02	Attendance: one table and one dialog, "Full attendance" is one button. Evaluation: one page with the rubric and the comment	Met

Repeated from Increment 1 and still met: CI runs every check above on the pull request (NFR-MNT-01); the OpenAPI document and the generated types match the code (NFR-MNT-02); the new settings are editable on the settings page (NFR-MNT-03); `docker compose up -d --build` starts the stack with the model (NFR-PRT-02); new pages have no horizontal scrolling at 375 px (NFR-USA-03).

5. AI Evaluation

Run on Oct 26, 2026 with `ai-eval/evaluate.py`. The set is synthetic (C-09): 20 student profiles (10 Arabic, 10 English) and 30 vacancies, written and labelled by the author (decision RQ-4).

Measure	With the model	Rules only (`--basic`)	Random order	Target
Skill extraction recall	97.2% (English 97.4%, Arabic 96.9%)	same (no model involved)	–	≥ 70%
Relevant vacancy in the top 5	100%	100%	79.7%	≥ 80%
Relevant vacancy first	100%	100%	37.4%	–
Mean reciprocal rank	1.000	1.000	0.559	–
Text similarity alone, top 5 / first	100% / 100%	–	–	–

How to read these numbers:

- **The top-5 target is met, but on this set it shows little.** A profile has 9.2 eligible vacancies on average, 2 or 3 of them relevant, so a random order already puts one in the top 5 for 79.7% of the profiles. The ranking's real strength shows in the other two rows: the relevant vacancy comes first for every profile, against 37.4% by chance.
- **The model and the rules agree here.** Text similarity alone also ranks a relevant vacancy first for every profile, so the model works; the set is simply too easy to show what it adds to the skill rules.
- **The recall figure is optimistic.** Arabic recall was 67.7% at the first run. The fixes (attached particles and six synonyms, Section 7) were found on this same set, so it is no longer a held-out measure.
- **Missed skills:** Machine Learning (E03), ERP Systems (E09), Project Management (A07), GitHub (A08).

A harder held-out set (more eligible vacancies per profile, vacancies described only in free text, profiles labelled by another person) is planned for M6.

6. M4 Demo Scenario

`test_m4_demo_scenario` runs steps 1–9 of 01_Requirements Section 7 through the HTTP API, with the clock moved 11 days for steps 3 and 4:

Step	Result
1. Student dashboard; recommended vacancies with fit, matched and missing skills, Advisory; development plan; applies	Pass
2. Model off: the same list is labelled basic	Pass
3. Company dashboard: applicants to review and overdue attendance; full week and partial week with a note	Pass
4. Week older than 7 days refused; coordinator unlocks; supervisor corrects it	Pass
5. Periodic evaluation visible to the student at once; final evaluation hidden until Final	Pass
6. Grade submission refused with the missing items; components entered; company part computed (80.00); total 86.00 submitted	Pass
7. One grade returned with a comment and resubmitted; two grades approved in bulk; the student sees the Final grade and the final evaluation	Pass
8. Final grade reopened with a reason; the audit entry carries it	Pass
9. Cycle statistics (2 placed, attendance 80%) and platform figures (counts only)	Pass
10. Evaluation script run	Pass (Section 5)

The same screens were walked through in the browser on the demo data in both languages and at phone width (Section 4).

7. Defects Found and Fixed

Defect	Found by	Fix
A week past its edit window but never recorded was treated as locked and left out of the company's overdue count	`test_at_att_04…`	State and lock reported separately
Arabic skill extraction recall 67.7%	Evaluation script	Attached particles (و، ف، ب، ل، ك، لل) handled; six synonyms added, and brought to existing databases by migration `0004`
The model version showed `@local`	Evaluation script	Read from the cached snapshot
The web app showed the bare status text ("Conflict") instead of the server's message on every typed call, since Increment 1	Browser check (grade submission)	Message taken from the body the client already read; `client.test.ts`
The first ranking after a restart waited about 90 s for the model, which also contacted the Hugging Face Hub	Browser check	Model loaded at start in the background, from the local copy
Reading a 5 MB PDF packed with text took 89 s, and a compressed file could unpack to megabytes	New timed test of NFR-PER-03	Text read from at most 20 pages, 512 KB of instructions per page, 2 MB per file; DOCX above 50 MB unzipped refused for reading
The API image stopped at start outside Docker Compose: its default file folder was not writable	Container started as on the Space	Image default `LOCAL_STORAGE_DIR=/data/storage`

Defect	Found by	Fix
The API could start in production with the development signing key	Deployment review	Refused at start
Wording and display (outdated texts on two pages, "Grade: Draft" shown to the student, a submitted evaluation shown as "awaiting review", twelve precondition lines, "3 طالب", no Cancel in the attendance dialog, dashboard grade links to the wrong page)	Browser check	Fixed; listed in 02_Design.md Section 12

8. Deployment Checks

The reference deployment is prepared but not switched on: it needs the owner's Hugging Face account and a Gmail app password (Deployment.md, Section 3). What was checked without it:

- **Image:** built with the model (about 3 GB), started with `ENV=production`, `PORT=7860`, and the scheduler on, against the local database. The model loaded in about 15 s with no network call, the deep health check answered with the database and the model, the development routes were absent, and the Missing and embedding jobs ran.
- **Storage adapter, bucket copy, and restore:** tested against a stand-in of the Supabase Storage API (`test_storage.py`).
- **Deploy script:** what it uploads and the Space address (`test_deploy.py`); the workflow does nothing while its secrets are missing.
- **Database backup:** dump with `pg_dump` 17, encryption and decryption with `gpg`, restore into a scratch database. Every table had the same row count, embeddings included.
- **Web build:** with `VITE_API_URL` set, every call goes to that address.

9. Known Limitations and Follow-Up

- **Deployment switch-on:** waits for the owner's steps. The Space must be public, so the backend code is visible there (not the secrets). NFR-PER-01 and NFR-REL-01 are measured on it in M6, as planned.
- **AI evaluation:** the set is easy and partly tuned on itself (Section 5); a harder held-out set is planned for M6.
- **Stretch follow-ups from Increment 1 not reached:**
  - **Tailwind CSS 4:** `npm audit` still reports 3 moderate advisories in Tailwind 3's build tools, which never reach users.
  - **Backend type checker:** not added.
  - **Automated browser tests:** not added; the browser checks were done by hand.
- **Registration limit behind the Space's proxy:** with `FORWARDED_ALLOW_IPS=*` the visitor's address comes from the forwarded header, which a client could forge to avoid the limit. Accepted for the demo; narrowed if abused.
- **First ranking after a start:** the model takes about 15 s to load in the container; a ranking requested in that time waits for it.
- **Email:** tested through Mailpit; Gmail delivery is checked at switch-on.

10. Definition of Done

Item (01_Requirements Section 8)	Status
Code merged into `main` through a pull request with CI green	Pull request opened from `increment-2`; merge after CI passes

Item (01_Requirements Section 8)	Status
Acceptance tests automated and passing; endpoints in the permission matrix	Done (Sections 3 and 1)
Screens in Arabic and English following the wireframes	Done (browser check, Section 4)
Actions write audit entries	Done: attendance records and unlocks, evaluation submit and reopen, grade update, submit, return, approve, and reopen (`test_m4_demo_scenario`, AT-WFL-07)
Design, API, or SRS updated where the implementation changed something	Done: CR-08, Design 1.4, API 1.3, SRS 1.5, SPMP 1.4, 02_Design.md Section 12
Demo scenario runs without help	Done (Section 6)
Test report with the evaluation script's numbers	This document

Increment 3 – Test Report

Status: Complete · Oct 27, 2026 · Author: Salam Alsaour
Input: 01_Requirements.md (acceptance tests, Section 6; demo scenario, Section 7; Definition of Done, Section 8), 02_Design.md (test design, Section 11; changes, Section 12)
Build tested: branch `increment-3`; backend Python 3.11, PostgreSQL 16 with pgvector 0.8.7; embedding model `intfloat/multilingual-e5-small@614241f622f5`; frontend Node 22, React 18, Vite 8

1. Summary

Check	Result
Backend tests (pytest, real PostgreSQL)	**187 passed**, 0 failed (158 at the end of Increment 2)
Backend line coverage	**95%** of `app` (gate 70%)
Permission matrix	145 routes listed (16 new); **696** calls by callers that are not allowed, all answered `401` or `403`
Acceptance tests of 01_Requirements Section 6	**21 of 21 pass in their automated part**; the device parts of AT-PWA-01 – 03 (install on a phone, a real push, airplane mode) are planned for M6 (Section 9)
Non-functional checks of 01_Requirements Section 4.2	**11 of 11 met**
Similarity evaluation (`ai-eval/evaluate.py`)	**10 of 10** copied reports flagged (target 9 or more), **0 of 10** original reports flagged (target 1 or fewer); the copied reports hold 44–54% copied text. The setting was chosen on this set (Section 5)
Frontend	TypeScript strict build clean; **11 Vitest tests passed** (translation keys, date format, API errors, installable app); production build succeeds; 745 translation keys in each language
Lint	`ruff` clean on `app`, `tests`, `alembic`, `scripts`
Dependency audits	`pip-audit` on `requirements.txt`: no known vulnerabilities; `npm audit --omit=dev`: 0 vulnerabilities. Both now run in CI
M5 demo scenario	Steps 1–5 and 7–9 automated end to end through the HTTP API (`test_m5_demo_scenario`); step 6 checked in the browser; step 10 run
Container	The API image builds with the new libraries; inside it, the Arabic grade sheet is produced with both fonts embedded
Committed scope	17 of 17 requirements (all Should), FR-ORG-07 included; the dashboard items left by Increment 2; the dependency audit. The three stretch follow-ups were not reached again (Section 9)

2. How to Repeat the Tests

```
docker compose up -d db mailpit
cd backend
pip install -r requirements-dev.txt
ruff check app tests alembic scripts
pytest --cov=app                          # 187 tests; the tests use a stand-in for the model
pip-audit -r requirements.txt
cd ../frontend
npm ci
npx tsc -b && npx vitest run && npx vite build
npm audit --omit=dev
```

The evaluation needs the model packages (`pip install -r backend/requirements-ai.txt`):

```
python ai-eval/evaluate.py
```

The installable app and the offline page: `npm run build` then `npm run preview` in `frontend/` (http://localhost:4173), with the API running.

3. Acceptance Tests

Test	Automated test(s), or check	Result
AT-SIM-01	`test_at_sim_01_02_copied_report_flagged_within_the_university_only` (the same text stored at another university is never a source), `test_at_sim_01_earlier_versions_of_the_same_report_are_not_sources`	Pass
AT-SIM-02	The same API test (score, cycle, week, passages; the author only for the author's coordinator); evaluation script 10 of 10 and 0 of 10 (Section 5)	Pass
AT-SIM-03	`test_at_sim_03_advisory_decision_changes_nothing_else` (comment required, only a flagged check, status unchanged, audited)	Pass
AT-NOT-01	`test_at_not_01_list_unread_count_and_one_email_each`, `test_failed_email_is_retried_then_marked_failed`	Pass
AT-NOT-02	`test_at_not_02_reminder_once_between_24_and_23_hours`	Pass
AT-NOT-03	`test_at_not_03_missing_notice_to_student_and_supervisor`	Pass
AT-NOT-04	`test_at_not_04_escalation_on_day_4_never_twice` (reports, attendance, evaluations; the list and the dashboard count)	Pass
AT-NOT-05	`test_at_not_05_application_and_placement_events`, `test_at_not_05_report_evaluation_and_grade_events`, `test_at_not_05_verification_events`; the transfer events in AT-ORG-07	Pass
AT-NOT-06	`test_at_not_01…` (an Arabic and an English recipient of the same events); M5 step 4	Pass
AT-EXP-01	`test_at_exp_01_grade_sheet_values_equal_stored_values` (every cell read back from the Excel file; the PDF text read back), `test_unit_formula_injection_guard`	Pass
AT-EXP-02	`test_at_exp_02_term_summary_equals_the_dashboard`	Pass
AT-EXP-03	`test_at_exp_03_training_record_for_the_assigned_supervisor` (another supervisor and the student are refused)	Pass
AT-PWA-01	`src/pwa.test.ts` (manifest fields and icons, service worker rules, page links); production build in the browser: service worker active, app shell cached, manifest served	Pass; the install itself on phones in M6

Test	Automated test(s), or check	Result
AT-PWA-02	`test_at_pwa_02_push_has_title_and_link_only` (payload `{title, link}` only, email when no subscription), `test_gone_subscription_is_deleted`, `test_push_subscription_endpoints`; the page shows the right state when the browser blocks notifications	Pass; a real push on a device in M6
AT-PWA-03	Production build with the browser's offline state: the offline banner appears and every action button is disabled; back online, the banner goes	Pass; airplane mode on a phone in M6
AT-AUD-03	`test_at_aud_03_history_matches_the_audit_entries`; the student never sees the similarity decision (`test_at_sim_03…`)	Pass
AT-ORG-07	`test_at_org_07_coordinator_role_transfer`, `test_at_org_07_company_admin_transfer`	Pass
AT-PER-04	`test_at_per_04_result_within_2_minutes_with_10000_stored_reports` (Section 4)	Pass
AT-PER-05	`test_at_per_05_grade_sheet_of_100_students_under_60_seconds` (Section 4)	Pass
AT-REL-03	Every job test runs the job twice and checks that nothing is sent twice; the escalation test jumps the clock as after a downtime and the next run catches up	Pass
AT-L10N-03	`test_at_l10n_03_arabic_pdf_embeds_the_arabic_font`; the exported pages rendered and checked by eye (Section 7)	Pass

4. Non-Functional Checks

ID	Evidence	Result
NFR-PER-04	With 10,000 stored reports (20,000 passages in the HNSW index), submitting a report and running its check took **0.35–0.42 s** over two runs with the stand-in model (limit 2 minutes). With the real model, embedding a report's passages adds well under a second	Met
NFR-PER-05	100-student grade sheet: **1.0–1.3 s** for Excel and **2.4–4.6 s** for PDF over two runs, the slower one while the evaluation ran beside it (limit 60 s each)	Met
NFR-REL-03	Jobs every 1 to 15 minutes; idempotency keys per recipient; repeated runs send nothing twice (AT-REL-03)	Met
NFR-PRV-03	The push payload is the title and the link only, with no name or grade (`test_at_pwa_02…`, M5 step 5)	Met
NFR-L10N-03	fpdf2 with HarfBuzz shaping and the embedded Noto Naskh Arabic font; letters joined and right to left in the rendered pages, also in the Linux container	Met
NFR-L10N-04	The similarity set has 5 Arabic and 5 English pairs of copied and original reports, several Arabic ones with English terms inside; all copied ones flagged, no original flagged	Met
NFR-AI-01	The similarity panel is under the Advisory label with its score and passages; a decision changes no status (AT-SIM-03)	Met
NFR-AI-02	Each check stores `model_version` and the threshold in force	Met

ID	Evidence	Result
NFR-AI-03	`ai-eval/data/similarity.json` and the script; two runs gave the same numbers	Met
NFR-SEC-05	Dependency audits clean and in CI; the ZAP scan is part of the release check in M6	Met for this increment
NFR-L10N-01	Browser check of every new screen in Arabic and English: notification list and bell, similarity panel and list, history panels, escalations, exports, transfer dialog, offline banner	Met

New pages have no horizontal scrolling at 375 px (NFR-USA-03). The checks of Increments 1 and 2 still pass in CI.

5. Similarity Evaluation

Run on Oct 27, 2026 with `ai-eval/evaluate.py`. The set is synthetic (C-09), written and labelled by the author (Increment 3, RQ-6): 10 stored reports, 10 reports that copy two or three of their sentences (44–54% of the copying report, some with a word changed) among their own sentences, and 10 original reports on the same topics as the stored ones, half in Arabic and half in English.

Measure	Result	Target
Copied reports flagged	10 of 10 (English 5, Arabic 5)	9 or more
Original reports flagged	0 of 10	1 or fewer
Scores of the copied reports	0.42 – 0.54 (close to their real copied share)	–
Scores of the original reports	0.00	–

How to read these numbers:

- **The first version of the method failed this set.** With passages of 40–80 words (the design), only 4 of 10 copied reports were flagged and no Arabic one, because the copied sentences shared a passage with the student's own words. Sentence-sized passages, compared with the source's nearby passages taken together, give the result above (CR-09).
- **The setting was chosen on the same set.** The numbers are therefore optimistic. A held-out set, written by someone else, is planned for M6 together with the matching one.
- **The first version of the set was too easy.** Its copied reports held over 70% copied text, which hid the problem; the set now follows the 40% of the acceptance criterion.

The matching and skill-extraction numbers of Increment 2 are unchanged (skill recall 97.2%; relevant vacancy first for every profile, against 37.4% by chance).

6. M5 Demo Scenario

`test_m5_demo_scenario` runs these steps through the HTTP API:

Step	Result
1. A weekly report copied from a report of an earlier cycle is flagged; the supervisor's dashboard shows it; score, source cycle and week, passages	Pass
2. "Overlap confirmed" with a comment; the report status unchanged; the coordinator sees the author	Pass
3. The same text at another university is not a source	Pass
4. Reminder, Missing notice (student and supervisor), escalation on day 4; emails in Arabic and English	Pass
5. A Final grade: in-app notification, email, and web push for the student	Pass

Step	Result
6. Installable app; offline banner and disabled buttons	Pass in the browser (Section 3)
7. Grade sheet (Excel and PDF) with Arabic names, term summary, training record	Pass
8. History of a report, an evaluation, and a grade	Pass
9. Coordinator role transferred; the previous coordinator loses access; audit entry	Pass
10. Evaluation script	Pass (Section 5)

In the browser, on the seeded demo data:

- The demo student submitted the former student's paragraph, and the academic supervisor saw the flag on the dashboard and the three copied sentences side by side.
- The supervisor recorded the decision, and it appeared in the report's history.
- The coordinator's escalations, exports, and notifications were opened in Arabic and English and at phone width.
- The Platform Admin's transfer dialog listed the right candidates.

7. Defects Found and Fixed

Defect	Found by	Fix
Half-copied reports were missed (4 of 10 found)	Evaluation script, once the set followed the 40% criterion	Sentence-sized passages and the joined-neighbour rule (CR-09)
A copied passage that spans two source passages was missed	Evaluation script (report C07)	Compared with the source's nearby passages taken together
The side-by-side view listed unrelated source passages	Browser check	Only passages that share words with the copied one
Grade weights printed as "30.00"; Arabic footer in the wrong order; a gap left by the total page count; percent sign placement; cycle name and dates reordering each other	Rendering the exported PDFs	Weight without decimals; footer in separate cells with the page number only; Arabic percent sign; dates on their own line
The history arrow pointed backwards in English	Browser check	Arrow by direction
"11 أيام" in Arabic	Browser check	Days overdue shown as a number under its header
The escalation filter was named `kind` instead of the specification's `type`	Review against the API specification	Renamed
The first similarity speed test loaded its 20,000 passages row by row and ran for minutes	Running the test	The data is made inside the database

8. Notes on the Tests

- **Background check in tests:** the test client runs the background task of a submission before it returns, so the API tests see the finished check, as a user does a moment after submitting.
- **Model in tests:** the tests use the stand-in embedder of Increment 2. The real model was used for the evaluation, the demo data, and the browser check.

9. Known Limitations and Follow-Up

- **Device checks (M6):**

    - Installing the app from Chrome on Android and Safari on iOS.
    - Receiving a real web push: the built-in browser used here blocks notifications.
    - Opening the app in airplane mode: the offline state was simulated in the browser here, and the service worker's offline page was checked by its cache contents.
- **Similarity:**
    - The setting was tuned on its own evaluation set (Section 5).
    - The check needs the model. Without it, checks wait and the retry job runs them, up to 3 attempts.
- **Stretch follow-ups not reached again:** Tailwind CSS 4, a backend type checker, and automated browser tests. `npm audit` still lists 3 moderate advisories in Tailwind 3's build tools, which never reach users.
- **Notifications:**
    - Delivered by a job every minute, so an email arrives within about a minute of the event.
    - Real Gmail delivery is checked when the deployment is switched on.
- **PDF page total:** pages show their number but not the total.

10. Definition of Done

Item (01_Requirements Section 8)	Status
Code merged into `main` through a pull request with CI green	Pull request opened from `increment-3`; merge after CI passes
Acceptance tests automated and passing, or checked in the browser where marked; endpoints in the permission matrix	Done (Sections 3 and 1); device parts in M6
Screens in Arabic and English following the wireframes	Done
Actions write audit entries	Done: similarity decision, exports, role transfers; the history shows them
Design, API, or SRS updated where the implementation changed something	Done: CR-09, Design 1.5, API 1.4, SPMP 1.5, Deployment 1.1, 02_Design.md Section 12
Demo scenario runs without help	Done (Section 6)
Test report with the similarity evaluation	This document

Increment 4 – Test Report

Date: Oct 28, 2026 · Author: Salam Alsaour
Scope: 01_Requirements.md (7 Could requirements, FR-PWA-03 as changed by CR-10, and the release checks of Section 4.2) · **Design:** 02_Design.md
Milestone: M6 – Testing complete (Dec 10, 2026). The build and the release checks are done; the user acceptance testing is prepared (05_UAT_Plan.md) and its results will be added here.

1. Summary

Check	Result
Backend tests (pytest, real PostgreSQL)	**201 passed**, 0 failed (187 at the end of Increment 3)
Backend line coverage	**95%** of `app` (7,087 statements; gate 70%)
Permission matrix	163 routes listed (18 new); **810** calls by callers that are not allowed, all answered `401` or `403`
Acceptance tests of 01_Requirements Section 6	**12 of 13 met**. AT-AI-03 is met for matching and not for similarity: 7 of 10 copied reports of the held-out set flagged, target 9 (Section 5)
Browser tests (Playwright)	**34 passed** on Chromium, WebKit, a Pixel 7, and an iPhone 14; 18 skipped by design (the layout checks run once). Firefox does not start on the development laptop, which blocks its executable, so it runs in CI
Frontend	TypeScript strict build clean; **19 Vitest tests passed**; production build succeeds; 839 translation keys in each language
Security scan (OWASP ZAP)	Web app baseline: 0 high, 0 failures. API scan in safe mode: 0 high, 0 warnings after two fixes (Section 6)
Load test (Locust, 50 users, 5 minutes)	7,785 requests, **0 errors**; 95th percentile **0.21 s** (limit 2 s)
Dependency audits	pip-audit and `npm audit --omit=dev`: no known vulnerabilities
AI evaluation, held-out sets	Matching top-5 **100%** (random order 35.7%); similarity **7 of 10** copied flagged, **0 of 10** originals (Section 5)
AI evaluation, tuning sets (unchanged)	Skill recall 97.2%; matching top-5 100%; similarity 10 of 10 and 0 of 10
Backup and restore	Dump of the check database restored into a new database; every table, the new ones included, has the same row count
M6 demo scenario	Steps 1–5 automated through the API on the demo data (`test_m6_demo_scenario`); step 6 in the browser tests; step 7 is Key_Results.md
Lint	`ruff` clean on `app`, `tests`, `alembic`, `scripts`, `loadtest`
Committed scope	8 of 8 requirements delivered

2. How to Repeat the Tests

```
cd backend && pytest --cov=app                                  # 201 tests
backend/.venv/Scripts/python ai-eval/evaluate.py --heldout      # from the repository root
```

The browser tests, the load test, and the ZAP scan run against the Docker stack on its own database, as the README shows (`docker-compose.check.yml`).

3. Acceptance Tests

Test	How	Result
AT-CER-01	`test_at_cer_01_only_a_final_grade_gets_a_certificate`: refused before Final; 26-character code; the PDF text holds the code, the name, and the company and no grade; the QR code read back from the PDF (zxing) equals the page address; a second valid one refused; the file only for the student and the coordinator	Pass
AT-CER-02	`test_at_cer_02_public_check_valid_revoked_and_altered` (fields exactly as FR-CER-02 lists; dashes and lower case accepted); browser test `public.spec.ts` on four engines	Pass
AT-CER-03	`test_at_cer_03_revoke_needs_a_reason_then_a_new_one_can_be_issued`	Pass
AT-EVL-05	`test_at_evl_05_…`: opens at the training end date (and with a Final grade, `test_feedback_opens_with_the_final_grade…`), once only, the coordinator sees it and the averages, and the company supervisor, Company Admin, academic supervisor, another coordinator, another student, and the Platform Admin are refused on every path	Pass
AT-DSH-03	`test_at_dsh_03_job_offer_rate_is_yes_over_answers`: 3 Yes of 4 answers is 75.0%; a student who skipped is not counted; Pending changed to Yes	Pass
AT-DSH-04	`test_at_dsh_04_leniency_alert_from_five_final_evaluations` (4.8 against 3.6 alerts) and `…four_evaluations_raise_no_alert`	Pass
AT-PRV-04	`test_at_prv_04_anonymize_after_the_retention_period`: refused while open and before the period; afterwards no name, email, student number, CV file, report text, comment, or certificate file can be retrieved; the audit log is masked; the student still in another open cycle is skipped; the four statistics answers are identical before and after; the cycle cannot be reopened. The demo data repeats it (`test_m6_demo_scenario`)	Pass
AT-PWA-03	Unit tests `offline.test.ts` (what the copy keeps and leaves out, another user, sign-out, idle time, draft expiry); browser test `offline.spec.ts` on four engines: saved copy with its time, the not-saved notice, the draft typed offline saved after reconnecting, nothing left after sign-out	Pass
AT-SEC-05	ZAP baseline and API scan (Section 6); dependency audits	Pass
AT-PER-01	Locust, 50 users (Section 4)	Pass on the local stack
AT-PRT-01	Browser tests on Chromium, WebKit, Pixel 7, iPhone 14; no horizontal scrolling at 360, 768, 1366, and 1920 px	Pass; Firefox in CI, real phones in the UAT
AT-USA-04	axe contrast check on 12 pages; keyboard-only browser test (sign in, open the reports, save a draft)	Pass
AT-AI-03	Held-out sets run once (Section 5)	Matching met; similarity **not met**

4. Release Checks (Section 4.2)

ID	Result	Status
NFR-SEC-05	ZAP: 0 high-risk findings (Section 6); audits clean	Met
NFR-PER-01	Local Docker stack on the development laptop, 50 users, spawn 5 per second, 5 minutes: 7,785 requests, 0 failures; 95th percentile 0.21 s over all requests. By kind: dashboard 0.14 s, placement list 0.52 s (the slowest), report slots 0.13 s, attendance 0.19 s, vacancies 0.29 s, draft save 0.20 s, sign-in 0.83 s (Argon2 on purpose). Slowest single request 1.9 s	Met on the local stack; repeated on the reference deployment after the switch-on
NFR-PRT-01	Browser tests (Section 3)	Met for four engines and sizes; Firefox in CI; phones in the UAT
NFR-USA-03	No horizontal scrolling at the four widths on 11 pages (sign-in, five of the student, five of the coordinator)	Met
NFR-USA-04	No contrast failure on 12 pages after one fix (Section 7); the main task done with the keyboard alone	Met
NFR-AI-03	Held-out sets in the repository, the script reproduces the numbers	Met; the similarity target is not (Section 5)
NFR-REL-02	Restore of a dump of the check database: identical row counts in every table, `certificates`, `company_evaluations`, and `outcome_surveys` included	Met
NFR-USA-01	UAT plan, task cards, consent text, and result sheet ready	Pending: run by the owner

5. Held-out AI Evaluation (RQ-6)

The sets were written on Oct 28, after the settings of CR-09 were frozen, and labelled before the first run. They ran once; no setting was changed afterwards.

Set	Result	Random order	Target
Matching: 10 profiles, 15 vacancies with one listed skill each, all open to every profile	Top-5 **100%**; top-1 **60%**; mean reciprocal rank **0.767**	35.7%; 7.3%; 0.232	80% top-5
Similarity: 4 sources, 10 copied reports (48–78% copied), 10 originals on the same topics	**7 of 10** copied flagged: all 7 copied word for word, none of the 3 reworded (HC04, HC05, HC10, score 0); **0 of 10** originals flagged	–	9 of 10; at most 1

What this shows:

- **Matching** holds on a harder set: the model ranks the right vacancy high even when the vacancy names one skill only.
- **Similarity** finds copied text reliably and never accused an original report, but a report reworded sentence by sentence escapes the word 5-gram rule entirely.
- **The 9-of-10 target** (O5) was written for "copied" reports. It is met for text copied word for word (7 of 7 here, 10 of 10 on the tuning set) and not for paraphrase.
- **The flag is advisory** and the supervisor decides (NFR-AI-01). The test report records the limit instead of retuning the setting on this set, which would make it a tuning set again.

- **Possible remedy:** a rule on the embedding similarity alone. It would also flag original reports on the same topic, so it needs its own evaluation. It is listed for after the project (Section 8).

6. Security Scan (NFR-SEC-05)

Target	Findings	Action
Web app (development server), ZAP baseline	Medium: no Content-Security-Policy, no anti-clickjacking header. Low: COEP, COOP, CORP, Permissions-Policy, `nosniff`. Informational only otherwise	The hosted app is served by Vercel with `vercel.json`. It already sent `nosniff`, `X-Frame-Options`, Referrer-Policy, and Permissions-Policy. **It sent no Content-Security-Policy although the design said so (defect 3)**: added with COOP. The production preview now sends the same headers, and a run in the browser showed no blocked resource
API, ZAP API scan in safe mode from the OpenAPI document	Low: `X-Content-Type-Options` and `Cross-Origin-Resource-Policy` missing on JSON answers	Both added to every answer (`test_every_answer_has_the_release_check_headers`); the rescan passed 120 rules with 0 warnings

The workflow `security-scan.yml` repeats both scans every week and fails on a high-risk alert.

7. Defects Found and Fixed

Defect	Found by	Fix
1. The server's 30-minute idle limit never ended a session while a window stayed open: the notification count asks the API every minute	Design review of the offline idle limit	The app counts idle time from the last key, click, or touch and ends the session itself, online and offline (CR-11)
2. The saved copy was written at most once a second, so a page opened just before a reload was missing offline	Browser test `offline.spec.ts`	The copy is also written when the page is left or reloaded
3. `vercel.json` sent no Content-Security-Policy, although Design 7.1 relies on one	ZAP baseline	Strict policy added. A browser run then showed that the build inlines small fonts as `data:` URLs, so `font-src` allows `data:`
4. API answers without `nosniff` and a resource policy	ZAP API scan	Middleware adds both
5. The demo clock buttons (development only) had a contrast of 3.86:1	axe check	White text
6. The registration audit entry stored the email address, which an anonymization could not remove from the append-only log	Design of NFR-PRV-04	The entry keeps the kind only; older entries are masked when read
7. Certificates listed in a changing order when two had the same time	Test with the frozen clock	The valid certificate first, then the newest

8. Limitations and Notes

- **Similarity and paraphrase:** Section 5. Students are told that the check is advisory; a supervisor reading the passages decides.
- **The held-out sets were written by the author**, like the tuning sets, because no second person was available (RQ-6). They are independent of the settings, not of the author.

- **The load test ran on the development laptop.** The free Space has 2 CPU cores, so the numbers there will be higher. The same script runs against it after the switch-on.
- **Offline copy and app restarts:** the copy lives as long as the window, so closing the installed app ends it. This is chosen for privacy on shared devices (RQ-2).
- **Firefox:** covered by CI on Linux. On the development laptop Windows blocks the downloaded Firefox build.
- **Docker Desktop on the development laptop** crashed twice during this increment, after being stopped by force. Leftover socket files blocked its start until they were moved aside. This was a laptop problem, not a project problem, but it is noted because the demo runs on Docker.

9. Pending for M6

Item	Owner	Plan
UAT with at least 8 participants (O8)	Project owner	05_UAT_Plan.md; results into this report
Device check on an Android phone and an iPhone	Project owner	Same sessions (UAT plan, Section 7)
Firefox browser tests	CI	First run on this pull request
NFR-PER-01 and ZAP on the reference deployment	After the switch-on	Same scripts

10. Definition of Done

Criterion	Status
Code merged through a pull request with CI green	On this pull request
Acceptance tests automated and passing, or checked in the browser; endpoints in the permission matrix	Done, except the similarity part of AT-AI-03 (Section 5)
Screens in Arabic and English, following the wireframes or the existing layout	Done (`Verify_Mobile`; the other screens follow the existing pages)
Actions audited	Issue, revoke, evaluation of the company, survey, retention, anonymization
Documents updated	SRS 1.6, Design 1.6, API 1.5, SPMP 1.6, Deployment 1.2, CR-10, CR-11, README, Key_Results
Release checks with results here and in Key_Results	Done
UAT materials ready	Done; the sessions are pending

Increment 5 – Test Report

Date: Oct 29, 2026 · Author: Salam Alsaour
Scope: 01_Requirements.md, deliverables D1–D7 and checks C-1 to C-7 · **Design:** 02_Design.md
Milestone: M7 – Final delivery (Dec 15, 2026)

1. Summary

Check	Result
C-1 Links	Every relative link and image in 32 maintained documents points to an existing file (`check_links.py`)
C-2 Traceability	125 SRS requirements: 90 with an automated test named after them, 32 with a recorded result, 3 open (they need people or the hosted system), 0 untraced. `trace.py --check` passes
C-3 Screenshots	The 16 screenshots of the user guide were taken by `guide.capture.ts` on fresh demo data, without error
C-4 Exports	11 PDF files written by `export_pdf.mjs`; the SRS 1.6 built in Word and PDF on the course template (Section 4)
C-5 Demo dry run	Every step of the demo script ran on fresh demo data, through the screenshot script and `test_m6_demo_scenario`. A timed rehearsal by the presenter remains for the week of the demonstration
C-6 Review	Each deliverable read against 01_Requirements. The numbers in the final report and the slides were compared with Key Results and are equal (Section 3)
C-7 Regression	Backend: **202 passed**, 95% coverage, ruff clean. Frontend: tsc clean, 19 Vitest tests passed. Browser tests: **34 passed** on Chromium, WebKit, Pixel 7, and iPhone 14, 18 skipped by design. CI runs them again with Firefox on the pull request
Deliverables	D1–D7 done; the presentation is a 14-slide deck with speaker notes (link), downloaded as PowerPoint or PDF from its page

2. Defects Found and Fixed

Defect	Found by	Fix
On phones the side menu filled the whole first screen of every page, so each page opened on the menu	Reviewing the phone screenshots of the user guide	The menu folds behind a "Main menu" button below 768 px and closes after a link is followed (Design 1.7, Section 8.2); the browser tests open it where they need it
FR-MAT-06 had a test that did not carry its acceptance-test ID, so the requirement looked untraced	`trace.py`	Test renamed `test_at_mat_06_…`

Defect	Found by	Fix
NFR-USA-05 (error messages next to the field, in the user's language, saying how to fix) had no test	`trace.py`	`test_at_usa_05_field_errors_name_the_field_and_the_fix_in_the_users_language`, in Arabic and English

3. Numbers Compared (C-6)

Number	Key Results	Final report	Slides
Backend tests	202	202	202
Refused permission checks / endpoints	810 / 163	810 / 163	810 / 163
Browser tests	34	34	34
Requirements traced	125 (90 / 32 / 3)	125 (90 / 32 / 3)	125, 3 open
Matching top 5, tuning and held-out	100%, 100%	100%, 100%	100%, 100%
Copied reports flagged, held-out	7 of 10	7 of 10	7 of 10
50 users, 95th percentile	0.21 s	0.21 s	0.21 s
High-risk security findings	0	0	0

4. Submission Files (C-4)

`node docs/tools/export_pdf.mjs` wrote the PDFs of `docs/Final_Submission/`:

- the final report;
- the SPMP 1.7, the design 1.7, and the API specification 1.5;
- the five test reports;
- the UAT plan;
- the user guide;
- Key Results;
- the requirements status;
- the change requests;
- the demo script.

The SRS 1.6 was built with its own scripts on the course template:

- `gen_docx.py` fills the template;
- Word builds the table of contents and the PDF.

5. Notes

- **The local demo data.** Once during this increment the screenshot script ran against the main demo database instead of its own. It set six demo accounts to English, which was reverted, and submitted a weekly report of `s1@iau.edu.sa`, which cannot be undone in the system. The demo script starts from fresh demo data, so this has no effect on the demonstration. The script must always run with the API on its own database, as the user guide's header and README say.
- **User acceptance testing** was postponed by the project owner on Oct 29 (RQ-6). NFR-USA-01 stays open, with the plan ready.

- **Firefox** browser tests run in CI; Windows blocks the downloaded Firefox build on the development laptop.

6. Definition of Done

Criterion	Status
D1–D7 delivered	Done
Checks C-1 to C-7	Done; the timed rehearsal of C-5 is left for the presenter
Documents updated (SPMP 1.7, Design 1.7, document index, README, Key Results)	Done
Merged through a pull request with CI green	On this pull request